

Emmanouil Giortamis

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Research Interests

My research interests lie in quantum systems, where I design compilers, OS mechanisms, and hardware-software co-designs to scale quantum computing toward fault-tolerant quantum computation. I leverage IRs, ISAs, and cross-stack optimizations to improve execution fidelity, programmability, and resource efficiency across the quantum software stack.

My work emphasizes High-Performance Quantum Computing (HPQC) by integrating quantum resources into existing cloud and HPC infrastructures, and by offloading the classical part of the quantum stack to specialized accelerators such as GPUs and FPGAs for real-time error mitigation and decoding.

Previously, I worked in the distributed systems area, specifically in distributed shared logs, hardware-assisted replication protocols, and the theory and practice of local reads in asynchronous replication protocols.

Keywords:

Quantum Software Systems, Hardware-Software Co-design, High-Performance Quantum Computing, (Real-time) Quantum Error Correction, Fault-Tolerant Quantum Computation

Education

- **Ph.D.** in Computer Science (Sept 2021 - Dec 2026 (Expected))
School of Computation, Information and Technology
Technical University of Munich, Germany
Thesis: Architecting the Quantum Systems Stack: From Cloud Orchestration to Hardware-Level Quantum Control
Advisor: Prof. Dr. Pramod Bhatotia
- **M.Sc.** in Computer Science (Sept 2019 - July 2021)
University of Crete, Greece
- **B.Sc.** in Computer Science (Sept 2015 - July 2019)
University of Crete, Greece

Honors and Awards

- **Best Student Paper Award Finalist**
Paper: “Qonductor: A Cloud Orchestrator for Quantum Computing”
ACM/IEEE Supercomputing Conference (SC) ’25.
- **Best Paper Award**
Paper: “Recipe: Hardware-Accelerated Replication Protocols”
ACM/IFIP International Middleware Conference ’25.
- **Best Poster Nominee**
Poster: “The LAW Behind ALRs: Redefining Crash-Tolerant Reads”
ACM European Conference on Computer Systems (EuroSys) ’25.

- **Distinction DEPROFOIT, University of Crete, Greece, Sept 2018**
Undergraduate teaching assistant based on overall grades.

Employment

Technical University of Munich (TUM), Germany, Sept 2021 - present

Researcher & Scientific Employee

- Responsibilities: conducting research, teaching assistant.

planqc, Munich, Germany, Nov 2026 - Apr 2027

Internship

- Responsibilities: developing compiler software for neutral atom quantum computers.

ICS-FORTH, Heraklion, Greece, July 2018 - Sept 2018

Research Internship

- Responsibilities: experimental analysis of large-scale graphs on multiprocessor architectures.

ICS-FORTH, Heraklion, Greece, July 2017 - Sept 2017

Research Internship

- Responsibilities: developing a concurrent, shared-page memory allocator in C.

Ph.D. Dissertation

Topic: Architecting the Quantum Systems Stack: From Cloud Orchestration to Hardware-Level Quantum Control

Supervisor: Prof. Dr. Pramod Bhatotia

Expected submission: Dec 2026

My dissertation builds the systems software infrastructure to scale quantum computing from NISQ toward fault-tolerant quantum computation. In my thesis, I designed four main systems: *QOS* introduces OS-level abstractions for managing quantum resources, improving fidelity and Quality-of-Service for end users, and resource efficiency for the quantum cloud provider. *Qonductor* extends this to hybrid quantum-classical resources, providing transparent hybrid application development and deployment on hybrid clusters. *MCMit* is a hardware-software co-design that couples ML-based qubit readout discrimination with software error mitigation to mitigate mid-circuit measurement errors. Finally, *Oraqle* examines qubit readout as a systems-level design decision by providing a systematic evaluation of ML-based qubit-state discriminators and their impact on quantum error correction code performance.

Publications based on my Ph.D. thesis:

- *QOS: Quantum Operating System* — OSDI '25
Source code: <https://github.com/TUM-DSE/QOS>
- *Qonductor: A Cloud Orchestrator for Quantum Computing* — SC '25
Source code: <https://github.com/manosgior/Qonductor-SC25>
- *MCMit: Hardware-Software Co-Design for Mid-Circuit Measurement Error Mitigation* — MICRO '26
Source code: <https://github.com/manosgior/MCMit>
- *Oraqle: Qubit Readout as a Systems-level Knob* — under submission at SIGMETRICS '27
Source code: <https://github.com/manosgior/Oraqle>

Publications

Conference publications:

1. MCMit: Hardware-Software Co-Design for Mid-Circuit Measurement Error Mitigation
Emmanouil Giortamis, Felix Gust, Aleksandra Świerkowska, Sandra Stankovic, Yanbin Chen, Innocenzo Fulginiti, Xiaorang Guo, Benjamin Lienhard, Martin Schulz, and Pramod Bhatotia
IEEE/ACM International Symposium on Microarchitecture (MICRO) '26
2. Multi-Stage Mamba-Based Architecture for Fast and Scalable Superconducting Qubit Readout
Luca Otting, Xiaorang Guo, Emmanouil Giortamis, Benjamin Lienhard, Pramod Bhatotia, and Martin Schulz
IEEE International Conference on Quantum Computing and Engineering (QCE) '26
3. qTPU: Hybrid Tensor Networks for Quantum-Classical Acceleration
Nathaniel Tornow, Emmanouil Giortamis, Dennis Sprockholt, Christian B. Mendl, and Pramod Bhatotia
Symposium on Operating Systems Design and Implementation (OSDI) '26
4. ECCentric: An Empirical Analysis of Quantum Error Correction Codes
Aleksandra Świerkowska, Jannik Pflieger, Emmanouil Giortamis, and Pramod Bhatotia
ACM Conference on Measurement and Modeling of Computer Systems (SIGMETRICS) '26
5. Qconductor: A Cloud Orchestrator for Quantum Computing
Emmanouil Giortamis, Francisco Romão, Nathaniel Tornow, Dmitry Lugovoy, and Pramod Bhatotia
The International Conference for High Performance Computing, Networking, Storage, and Analysis (SuperComputing / SC) '25;
Finalist–Best Student Paper Award
6. QOS: Quantum Operating System
Emmanouil Giortamis, Francisco Romão, Nathaniel Tornow, and Pramod Bhatotia
Symposium on Operating Systems Design and Implementation (OSDI) '25;
7. QVM: Quantum Gate Virtualization Machine
Nathaniel Tornow, Emmanouil Giortamis, and Pramod Bhatotia
Conference on Programming Language Design and Implementation (PLDI) '25;
8. Weaver: A Retargetable Compiler Framework for FPQA Quantum Architectures
Oğuzcan Kirmemiş, Francisco Romão*, Emmanouil Giortamis, and Pramod Bhatotia*
International Symposium on Code Generation and Optimization (CGO) '25;
**Equal Contribution*
9. Recipe: Hardware-Accelerated Replication Protocols
Dimitra Giantsidi, Emmanouil Giortamis, Julian Pritzi, Maurice Bailleu, Manos Kapritsos, and Pramod Bhatotia
ACM/IFIP International Middleware Conference (Middleware) '25;
Best Paper Award
10. The LAW theorem: Local Reads and Linearizable Asynchronous Replication
Antonios Katsarakis, Emmanouil Giortamis*, Vasilis Gavrielatos, Pramod Bhatotia, Aleksandar Dragojevic, Boris Grot, Vijay Nagarajan, and Panagiota Fatourou*
International Conference on Very Large Data Bases (VLDB) '25;
**Equal Contribution*

11. FlexLog: A Shared Log for Stateful Serverless Computing
Dimitra Giantsidi, Emmanouil Giortamis, Nathaniel Tornow, Florin Dinu, and Pramod Bhatotia
ACM High-Performance Parallel and Distributed Computing (HPDC) '23

Pre-prints:

1. QShell: An FPGA Shell and Real-time Control-Plane OS for Quantum Error Correction
Theofilos Augoustis, Peter Wegmann, Anubhav Panda, Emmanouil Giortamis, and Pramod Bhatotia
Under submission, 2026
2. Zero-G: A Pre-Decoder-Aware Decoder for Quantum Error Correction
Peter Wegmann, Theofilos Augoustis, Aleksandra Świerkowska, Emmanouil Giortamis, and Pramod Bhatotia
arXiv pre-print, 2026
3. Harvest: Resource-Aware Quantum Compilation for Magic State Protocols
Jannik Pflieger, Aleksandra Świerkowska, Emmanouil Giortamis, and Pramod Bhatotia
arXiv pre-print, 2026
4. Oracle: An Empirical Analysis of Qubit Readout and Discriminators in Quantum Error Correction
Emmanouil Giortamis, Aleksandra Świerkowska, Sandra Stankovic, Felix Gust, Benjamin Lienhard, and Pramod Bhatotia
arXiv pre-print, 2026
5. MultiQ: Multi-Programming Neutral Atom Quantum Architectures
Francisco Romão, Daniel Vonk, Emmanouil Giortamis, Dennis Sprokholt, and Pramod Bhatotia
arXiv pre-print, 2026
6. Chipmunk: A Fault-Tolerant Compiler for Chiplet Quantum Architectures
Peter Wegmann, Aleksandra Świerkowska, Emmanouil Giortamis, and Pramod Bhatotia
arXiv pre-print, 2026
7. AtomGuard: An Architecture-Centric ISA for Verified Compilation on Neutral Atom Quantum Architectures
Francisco Romão, Nathaniel Tornow, Emmanouil Giortamis, Dennis Sprokholt, and Pramod Bhatotia
arXiv pre-print, 2026

Posters and Talks:

Posters

- The LAW Behind ALRs: Redefining Crash-Tolerant Reads
Antonios Katsarakis, Emmanouil Giortamis*, Vasilis Gavrielatos, Pramod Bhatotia, Aleksandar Dragojevic, Boris Grot, Vijay Nagarajan, and Panagiota Fatourou*
EuroSys '25 (best poster nominee), EuroSys '24, EuroSys '23
**Equal Contribution*

Conference Talks

- Middleware 2025, Nashville, USA, December 2025
Recipe: Hardware-Accelerated Replication Protocols
- SC 2025, Saint Louis, USA, November 2025
Qonductor: A Cloud Orchestrator for Quantum Computing

- OSDI 2025, Boston, USA, July 2025
QOS: Quantum Operating System
- HPDC 2023, Orlando, USA, June 2023
FlexLog: A Shared Log for Stateful Serverless Computing

Invited Talks

- Software Systems for Quantum Computing Research of the Chair of Computer Systems
Invited talk at Cisco Quantum Research, November 2025
Online
- Qonductor: A Cloud Orchestrator for Quantum Computing
Invited talk at the Quantum Computing Applications and Systems (QCAS) Workshop, part of ICCAD '25, October 2025
Munich, Germany
- Software Systems for Neutral Atom Quantum Architectures
Invited talk at QuEra Computing, July 2025
Boston, MA, USA
- Software Systems for Quantum Computing
Invited talk at MIT by Prof. William D. Oliver, July 2025
Cambridge, MA, USA
- MCMit: Mid-circuit Measurement Error Mitigation
Invited talk at the University of Wisconsin-Madison by Prof. Swamit Tannu, June 2025
Online
- MCMit: Mid-circuit Measurement Error Mitigation
Invited talk at UT Austin by Prof. Poulami Das, April 2025
Online

Service

Workshop Chair, qStack: International Workshop on Quantum Systems, co-located with SOSP '26
 Program Committee member, European Conference on Computer Systems (EuroSys) '27
 Program Committee member, ACM/IEEE International Symposium on Code Generation and Optimization (CGO) '27
 Invited reviewer for the *Quantum* journal
 Invited reviewer for the *ACM Transactions on Architecture and Code Optimization* (TACO) journal
 Invited reviewer for the *IEEE Access* journal
 IEEE Quantum Week 2024, Student Volunteer

Open Source Projects

- QOS: Quantum Operating System
<https://github.com/TUM-DSE/QOS>
- Qonductor: A Cloud Orchestrator for Quantum Computing
<https://github.com/manosgior/Qonductor-SC25>

- MCMit: Hardware-Software Co-Design for Mid-Circuit Measurement Error Mitigation
<https://github.com/manosgior/MCMit>
- Oracle: An Empirical Analysis of Qubit Readout and Discriminators in Quantum Error Correction
<https://github.com/manosgior/Oracle>
- ECCentric: An Empirical Analysis of Quantum Error Correction Codes
<https://github.com/TUM-DSE/eccentric>
- qTPU: Hybrid Tensor Networks for Quantum-Classical Acceleration
<https://github.com/TUM-DSE/qtpu>
- QVM: Quantum Gate Virtualization Machine
<https://github.com/TUM-DSE/qvm>
- Weaver: A Retargetable Compiler Framework for FPQA Quantum Architectures
<https://github.com/TUM-DSE/weaver>
- FlexLog: A Shared Log for Stateful Serverless Computing
<https://github.com/TUM-DSE/FlexLog>

Teaching Experience

Teaching Assistant:

- Cloud Software Engineering lab, TU Munich, SS 22, WS 23-24, SS 24, SS 25 (lead), SS 26
- Quantum Software Systems seminar: TU Munich, SS 23 (lead), SS 24, WS 24-25, WS 25-26
- Distributed Systems lecture, TU Munich, WS 21-22, WS 22-23
- Languages and Compilers lecture, University of Crete, SS 21
- Introduction to Computer Science lecture, University of Crete, WS18-19, WS 20-21
- Principles of Distributed Computing lecture, University of Crete, SS 20
- Data Structures lecture, University of Crete, WS 19-20

Mentored Ph.D. Students:

- Real-time Control Planes for Quantum Error Correction
Anubhav Panda
Ph.D. student (2026 - present)
- Scalable Decoding Algorithms for Real-time Quantum Error Correction
Peter Wegmann
Ph.D. student (2026 - present)
- Compiler Integration of Quantum Error Correction Codes
Aleksandra Świerkowska
Ph.D. student (2024 - present)
- Systems Software for Neutral Atom Quantum Computers
Francisco Romão
Ph.D. student (2023 - present)

Supervised Theses:

- Optimizing Scheduling for Lattice Surgery with Magic State Cultivation
Jannik Pflieger
M.Sc. thesis
- Scalable Fault-Tolerant Quantum Compiler for Chiplet Architectures
Peter Wegmann
M.Sc. thesis
- Real-time and parallel task scheduling for Quantum Computing
Marcin Praski
M.Sc. thesis
- Hardware-aware Optimal Quantum Circuit Cutting and Knitting
Thang Tran
M.Sc. thesis
- Quantum Circuit Transpilation: Experimental Analysis and Subarchitecture Selection
Zeynep Erdogan
M.Sc. thesis
- Scalable Quantum Cloud Scheduling: Optimizing Resource Allocation for Efficient NISQ Computing
Dmitry Lugovoy
M.Sc. thesis
- Microservice Architecture in Practice: Debugging the Behaviour of Concurrent Applications at financial.com AG
Jonathan Ryan Wijaya Tumboimbela
M.Sc. thesis
- Extensions to QStack: Virtual Qubit Routing and SuperMarQ Benchmarks
Ahmed Darwish
Guided research
- A System Stack for Distributed Quantum Computing
Nathaniel Tornow
Guided research
- DQS: A Framework for Efficient Distributed Simulation of Large Quantum Circuits
Nathaniel Tornow
B.Sc. thesis

Skills

- **Languages:** C, Python (expert), Unix shell, C++ (competent);
- **Frameworks:** Qiskit, PennyLane, Cirq, OpenMP, MPI, NVIDIA cuQuantum, CUDA, LLVM, Slurm;
- **Containerization & Orchestration:** Docker, containerd, Kubernetes (K8s);
- **CI/CD:** Git;
- **Soft skills:** Technical/Scientific writing, Leading research projects, Presenting complex ideas to non-experts, Comfortable working under uncertainty (exploratory tech), Mentoring students.

References

- **Prof. Dr. Pramod Bhatotia**
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